AI Research Copilot: Evaluating Duplicate-Aware Evidence Aggregation for Selective Scientific Question Answering

Final Academic Project Report & Empirical Evaluation Study

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Abstract

Retrieval-augmented generation (RAG) systems in specialized scientific domains frequently encounter evidence insufficiency, where retrieved passages discuss relevant topics but omit exact parameters, causal definitions, or empirical mechanisms required to formulate a factual answer. Conventional systems rely on scalar cross-encoder relevance thresholds that fail to detect evidence gaps or redundant passage stuffing. This project investigates the research question: Does aggregating distinct evidence by paper and section, combined with exact within-paper passage deduplication, improve answer-or-abstain decisions over relevance-only baselines at matched coverage? We develop a controlled 120-family scientific benchmark (480 total contexts across 160 papers), rigorously partitioned into 60 training families, 24 validation families, and 36 held-out test families (144 test contexts across 36 paper clusters). All models (B0 Always Accept, B1 Max Relevance, B2 Historical Heuristic Gate, M1 Relevance Raw, M2 Relevance Dedup, M3 Full Structure Raw, M4 Full Structure Dedup, and a Length Diagnostic Shortcut) were fit strictly on training data, tuned and frozen on validation data, and evaluated on the held-out test set with zero parameter fitting. Models, scalers, and validation thresholds were completely frozen before the new v2 test construction, while fully disclosing historical test-paper exposure during an earlier invalid Stage 4 run prior to the v2 benchmark protocol.

On the primary evaluation partition of 108 controlled test contexts, the hypothesized superiority of M4 over M1 is NOT SUPPORTED: M4 achieved a discrete Area Under the Risk-Coverage curve (AURC) of 0.4400 compared to 0.4329 for M1, yielding an observed paired difference of +0.0071 (bootstrap 95% percentile interval [-0.0503, +0.0679] across 2,000 paired cluster resamples; negative favors M4). Because the confidence interval spans zero, the hypothesized improvement is not supported, and equivalence is not established. Baseline B1 (Max Relevance) achieved the lowest AURC (0.4049) on the controlled partition. Descriptive subgroup analysis on 11 multi-passage families revealed lower absolute AURC for deduplicated models (M4=0.3803 vs M1=0.6894), while on 36 natural retrieval contexts M4 exhibited the lowest observed AURC (0.0501), though with only three negative instances the natural bootstrap interval spans zero ([-0.1176, +0.0160]). Finally, exact deduplication confers a bounded duplicate-invariance property under explicitly stated operating assumptions, with an observed 0.000000 score shift under tested post-selection duplicate injection, whereas raw models experience over 0.31 score distortion. This report documents the software architecture, formal mathematics, verified literature comparisons, and negative empirical findings.

1. Introduction & Problem Statement

> **[CORE PROBLEM STATEMENT]** Retrieval-augmented scientific question-answering systems must decide whether retrieved passages contain enough evidence to answer a question. High relevance scores and repeated passages do not necessarily establish that all requested information is available. This project investigates whether exact deduplication and paper/section features improve evidence-sufficiency ranking over relevance-only baselines, while measuring acceptance coverage and the proportion of accepted contexts judged insufficient.

Scientific literature question answering is an intrinsically high-stakes task. Unlike general open-domain question answering, academic queries demand exact mathematical equations, chemical pathways, hyperparameters, and experimental outcomes. When presented with queries regarding state-of-the-art research, standard large language models (LLMs) frequently hallucinate convincing yet factually fabricated citations and parameters. Retrieval-Augmented Generation (RAG) mitigates parametric hallucination by conditioning generation on passages fetched dynamically from an external corpus. However, standard RAG architectures operate under an unconditional answering assumption: they attempt to synthesize a response regardless of whether the retrieved context contains the necessary evidentiary premises.

Two Evidentiary Failure Modes: In specialized scientific corpora, two pervasive phenomena undermine conventional relevance-driven retrieval:

1. **Topical Relevance vs. Evidence Sufficiency:** Cross-encoders like BAAI/bge-reranker-base assign high semantic relevance scores (e.g., >0.90) to passages that share domain concepts, entity names, or background framing with the query, even when the passage entirely lacks the critical operational formula or quantitative finding requested.
2. **Redundant Evidence & Duplicate Stuffing:** Academic papers frequently repeat high-level claims across the abstract, introduction, and conclusion. Standard top-k retrieval often populates the prompt window with near-identical or identical sentences from the same paper, creating an illusion of multi-passage corroboration while displacing complementary evidence from methodological or experimental sections.

System Implementation Context: The software deliverable of this project—the AI Research Copilot—implements an end-to-end scientific literature assistant. Its software architecture comprises a local PDF ingestion pipeline (Docling / pypdf with section regex parsing), a persistent vector store (ChromaDB with nomic-embed-text embeddings), a sparse lexical index (BM25Okapi), a cross-encoder reranking stage (BAAI/bge-reranker-base), an in-memory citation graph (NetworkX DiGraph with PyVis visualization), and an interactive multi-tab user interface (Streamlit). Generation is handled by a hosted Gemini 3.5 Flash Lite adapter.

> **[SYSTEM VS. EXPERIMENT BOUNDARY]** Clear Separation of System Features vs. Evaluated Components: The broader AI Research Copilot software application encompasses PDF ingestion, vector indexing, interactive Streamlit dashboards, and citation network visualization. However, the scientific experiment described in this report specifically isolates and evaluates the selective evidence-sufficiency gating mechanism (models M1–M4 and baselines) on frozen contexts. No claim of end-to-end hallucination reduction or answer accuracy improvement is made without a corresponding generation experiment.

2. Literature Review & Related Work

To establish rigorous academic context, we compare our project against three peer-reviewed and primary preprint publications that formalize retrieval evaluation, evidence gaps, and selective generation. Each primary source was verified directly from official archives.

Citation / Archive	Formal Paper Title	Authors & Venue	Core Contribution
Yan et al. [1] arXiv:2401.15884	Corrective Retrieval Augmented Generation (CRAG)	S. Yan, J. Gu, Y. Zhu, Z. Ling (EMNLP 2024)	Evaluates retrieval confidence via fine-tuned T5 evaluator to trigger document refinement or external web search.
Joren et al. [2] arXiv:2411.06037	Sufficient Context: A New Lens on Retrieval Augmented Generation Systems	H. Joren, J. Zhang, C. Ferng, et al. (ICLR 2025)	Separates context sufficiency from LLM reasoning; proposes selective generation using sufficiency classifiers.
Chen et al. [3] arXiv:2609.37469	Relevance Is Not Sufficient Evidence: Detecting Evidence Gaps Before Generation in RAG (RINSE)	S. Chen, P. Hua, Y. Xiao (Preprint 2026)	Detects pre-generation evidence gaps combining component coverage, answer evidence, and small LM sufficiency.

2.1 Detailed Primary Source Analysis

A. Corrective Retrieval Augmented Generation (CRAG; Yan et al., EMNLP 2024): Yan et al. [1] address the issue of retrievers fetching sub-optimal or irrelevant passages that mislead downstream generators.

Verified Method Details: CRAG trains a lightweight retrieval evaluator (fine-tuned T5-large) to score query-document relevance and establish a three-way decision rule using dual fixed scalar thresholds: (1) Correct (confidence >= upper threshold): retain internal documents and apply sentence-level decomposition and filtering; (2) Incorrect (confidence < lower threshold): discard retrieved documents entirely and trigger external web search; and (3) Ambiguous: combine decomposed internal passages with web search results. CRAG demonstrates substantial accuracy improvements on open-domain QA (PopQA, TriviaQA).

Problem Framing & Differences: CRAG frames retrieval evaluation primarily as deciding whether to accept, rewrite, or seek external replacement for retrieved web/open-domain passages. In contrast, our project investigates a closed scientific corpus setting where external web search is absent, focusing on the empirical question of whether aggregating within-paper section diversity and deduplicating redundant passages improves selective answer-or-abstain decisions over relevance-only baselines. We do not claim general novelty over CRAG.

B. Sufficient Context: A New Lens on RAG (Joren et al., ICLR 2025): Joren et al. [2] address a key distinction in RAG evaluation: separating retriever failure (insufficient evidence in context) from generator failure (incorrect reasoning over sufficient evidence). They define context sufficiency as an objective property—whether a context provides enough factual premises to logically entail an answer—and demonstrate that frontier LLMs (e.g., GPT-4o, Gemini 1.5 Pro) frequently hallucinate when presented with insufficient context rather than abstaining. Crucially, Joren et al. also propose selective generation using sufficiency information, training binary classifiers to enable answer-or-abstain decisions evaluated along risk-coverage trade-off curves.

Problem Framing & Differences: While Joren et al. formalize selective generation using sufficiency classifiers over concatenated input contexts, our project specifically investigates how passage-level evidence features behave under aggregation: evaluating whether explicitly modeling bibliographic structure (paper and section groupings) and passage deduplication improves selective gating performance compared to scalar relevance alone.

C. Relevance Is Not Sufficient Evidence (RINSE; Chen et al., 2026): Chen et al. [3] demonstrate that topical relevance and evidence sufficiency are fundamentally distinct dimensions: passages may achieve high semantic similarity scores by discussing query entities while omitting the precise quantitative parameters, definitions, or causal links required to formulate a complete answer. RINSE detects pre-generation evidence gaps by combining three components: (1) Question Component Coverage (verifying that each sub-clause/requirement of the query has corresponding passage support), (2) Passage-Level Answer Evidence (identifying direct entity/value answers within passages), and (3) a small language model's joint sufficiency judgment over the collected evidence.
Problem Framing & Differences: RINSE addresses evidence gaps through multi-stage linguistic decomposition and local language-model validation. In contrast, our empirical project evaluates whether lightweight structural heuristics (within-paper section diversity, second-passage reinforcement) combined with exact passage deduplication can improve selective evidence gating without invoking additional generative LLM inference passes.

> **[ACADEMIC SCOPE & NOVELTY DISCLAIMER]** Framing of Contribution: This project does not claim to faithfully re-implement the complete systems of CRAG, Sufficient Context, or RINSE. Rather, our contribution is framed strictly as an evaluated empirical system and controlled comparative benchmark investigating whether duplicate-aware passage aggregation and structural features improve selective question answering over relevance baselines in scientific literature. Publication novelty remains unestablished.

3. Mathematical Formulation

All data structures, scoring functions, feature transforms, and metrics described below are transcribed directly from the verified codebase in backend/selective/ and evaluation/run_benchmark_v2_test_eval.py.

3.1 Context Representation & Exact Deduplication

Passage Data Structure: Let a scientific question be denoted by query q. The candidate context provided to the gating model is an ordered sequence of K scientific passages: C = (p_1, p_2, ..., p_K). Each passage p_i is an immutable dataclass record containing:

- **Text:** Text string text_i representing the raw passage content.
- **Reranker Score:** Cross-encoder relevance score s_i in [0, 1], evaluated via BAAI/bge-reranker-base (sigmoid activation, fp32).
- **Source Metadata:** Bibliographic metadata: primary source paper identifier pid_i and section heading sec_i.
- **Token Count:** Recomputed token count tok_i, evaluated using the pinned cl100k_base tokenizer.
- **Hash Integrity:** Deterministic content hash h_i = SHA-256(normalize_whitespace(text_i)).

Deduplication Operation (dedup): Exact within-paper deduplication is defined over the multiset of passages. Two passages p_i and p_j are identical if and only if:
p_i equiv p_j <=> normalize_whitespace(p_i.text) == normalize_whitespace(p_j.text) and p_i.pid == p_j.pid

Whitespace normalization collapses consecutive whitespace characters to a single space while strictly preserving case, acronyms, and mathematical/chemical notation. For each equivalence class of duplicate passages, the deduplication engine deterministically retains the single canonical representative possessing the maximum relevance score: p* = argmax_{p in [p_i]} p.s. The resulting deduplicated context is denoted C_dedup = (p'_1, ..., p'_{K'}), where K' <= K.

3.2 Feature Vector Extraction (M1–M4)

Feature Representation: To evaluate the 2x2 factorial experimental design (Relevance vs. Combined Structure; Raw vs. Deduplicated Passages), features are extracted either from the raw context C (K passages) or the deduplicated context C_dedup (K' passages):

1. Relevance Features (4D): Phi_rel(C) = [s_(1), s_(2), s_bar, s_(1) - s_(2)]^T in R^4, where s_(1) is the maximum reranker score, s_(2) is the second-highest score (or s_(1) if K=1), s_bar = (1/K) sum s_i is the mean passage score, and s_(1) - s_(2) is the top-score spread.

2. Structural Features (4D): Phi_struct(C) = [s_top_paper_second, s_other, H_sec, s_bar_sec]^T in R^4, where:

• s_top_paper_second: Score of the second-highest passage originating from the paper that produced s_(1) (0.0 if only one passage from that paper exists).

• s_other: Maximum score among all passages originating from any paper OTHER than the top paper (0.0 if all passages belong to the top paper).

• H_sec: Normalized Shannon entropy of passage section assignments: H_sec = - sum_{j} p(sec_j) log(p(sec_j)) / log(max(2, |distinct_sections|)).

• s_bar_sec: Mean relevance score aggregated across distinct section groups.

Model Feature Spaces: The four models of the 2x2 factorial grid are defined as:

• M1 (Relevance Raw): x = Phi_rel(C) in R^4

• M2 (Relevance Dedup): x = Phi_rel(C_dedup) in R^4

• M3 (Full Structure Raw): x = [Phi_rel(C); Phi_struct(C)] in R^8

• M4 (Full Structure Dedup): x = [Phi_rel(C_dedup); Phi_struct(C_dedup)] in R^8

• Shortcut Diagnostic: x = [K, sum tok_i] in R^2 (passage count and total context tokens).

3.3 Standardization, Gating Score, and Decision Rule

Logistic Decision Rule: Feature standardization is fit strictly on the training partition: mu_train = E[x_train], sigma_train = Std[x_train]. Test instances are scaled using frozen training parameters: x_scaled = (x - mu_train) / sigma_train. The gating probability is computed via an L2-regularized logistic regression model:

P(Y_suff = 1 | C) = sigma(w^T x_scaled + b) = 1 / (1 + exp(-(w^T x_scaled + b)))

Given an operating threshold T in [0, 1], the gating decision is binary: y_hat = 1 (Answer / Accept) if P(Y_suff = 1 | C) >= T, and y_hat = 0 (Abstain / Reject) if P(Y_suff = 1 | C) < T.

3.4 Evaluation Metrics & Tie-Correct Discrete AURC

Coverage and Risk: Let N denote sample size, y_i in {0, 1} ground truth sufficiency (1=sufficient, 0=insufficient), and y_hat_i in {0, 1} model decisions:

• Coverage: Cov = (1 / N) sum_{i=1}^N y_hat_i (fraction of contexts accepted).

• Evidence Selection Risk: Risk = (sum_{i=1}^N y_hat_i (1 - y_i)) / (sum_{i=1}^N y_hat_i). Crucially, if zero contexts are accepted (sum y_hat_i = 0), risk is mathematically undefined (None / null), NOT 0.0.

Discrete Expected AURC: Under uniform random tie-breaking, the discrete expected AURC is evaluated analytically across all accepted counts k in {1, ..., N}. Samples are partitioned into distinct score levels S_1 > S_2 > ... > S_M with group sizes n_m and error counts E_m = sum_{i in group_m} (1 - y_i). For

any integer count k lying in boundary group m (where cum_counts_{m-1} < k <= cum_counts_m):
Expected Errors(k) = cum_errors_{m-1} + (k - cum_counts_{m-1}) * (E_m / n_m)
Expected Risk(k) = Expected Errors(k) / k
Discrete Expected AURC = (1 / N) sum_{k=1}^N Expected Risk(k).
For an all-tied scorer, Expected Risk(k) is exactly E/N for all k, and discrete AURC equals the sample insufficient prevalence E/N. The insufficient-label prevalence represents the expected all-tied ranking AURC under the implemented convention, rather than an empirical lower bound or guessing floor.

3.5 Paired Cluster Bootstrap

Bootstrap Specification: To estimate uncertainty while respecting intra-paper and intra-family correlation, we implement a paired cluster bootstrap. Clusters are defined by primary source paper identifier pid. In each of B = 2,000 bootstrap iterations (random seed 20261104):
1. Sample M paper clusters with replacement: {pid*_1, ..., pid*_M}.
2. Gather all contexts associated with sampled clusters, preserving exact cluster multiplicity.
3. Evaluate all scoring models on the IDENTICAL resampled instances to compute paired differences: Delta_b = AURC_A(b) - AURC_B(b).
4. Compute the 95% percentile confidence interval: [q_0.025(Delta), q_0.975(Delta)]. Sign convention: Delta = AURC_A - AURC_B; negative values favor Method A.

3.6 Mathematical Duplicate-Invariance Argument

Proposition & Scope: We state the exact duplicate-invariance property and its rigorous operating assumptions:

> **[DUPLICATE INVARIANCE ARGUMENT & ASSUMPTIONS]** Duplicate-Invariance Property: Let C = (p_1, ..., p_K) be a fixed candidate context record. Suppose an exact duplicate passage p_dup is appended to C, where p_dup shares identical whitespace-normalized text, identical paper ID, identical section, and identical reranker score with an existing passage p_i in C. Under the following explicit assumptions:
> (A1) Fixed Context: The input candidate pool is fixed post-retrieval; no re-retrieval occurs.
> (A2) Metadata Equivalence: p_dup has identical paper_id, section, and score s_dup = s_i.
> (A3) Deterministic Canonical Representative: Within-paper deduplication collapses {p_i, p_dup} into p* = argmax(score) = p_i.
> (A4) Downstream Feature Dependency: Gating features Phi(C) are evaluated strictly on dedup(C).
> (A5) No Token Budget Displacement: Adding p_dup does not displace other distinct evidence passages under a context token cap.
> Then: dedup(C union {p_dup}) = dedup(C), which implies Phi(dedup(C union {p_dup})) = Phi(dedup(C)), and consequently:
> | Score_M4(C union {p_dup}) - Score_M4(C) | = 0.000000 (Bounded Duplicate Invariance under A1–A5).
>
> Separately, in empirical testing on 36 fixed test contexts with post-selection duplicate injection, M2, M4, and B1 exhibit an observed 0.000000 score shift, whereas raw models exhibit shifts of 0.318703 (M1) and 0.312393 (M3).
>
> Scope Limitations: This is not an unconstrained invariance theorem. Invariance does NOT hold if: (1) duplicates displace distinct passages during budget-constrained retrieval; (2) passages are paraphrases rather than exact string matches; (3) metadata or section labels differ; or (4) retrieval queries are re-executed.

4. Experimental Design & Benchmark Construction

Corpus Partitioning: The experimental evaluation is structured as a group-partitioned, 4-condition controlled benchmark across 160 eligible scientific papers (7,368 real text chunks) from arXiv. Development data (40 historical papers) was quarantined to prevent split contamination.

- **Training Partition:** 80 papers, 60 question families, 240 contexts (ctx_tr_0001 to ctx_tr_0240).
- **Validation Partition:** 32 papers, 24 question families, 96 contexts (ctx_val_0001 to ctx_val_0096).
- **Held-Out Test Partition:** 48 papers, 36 question families, 144 contexts (ctx_te_0001 to ctx_te_0144). Primary source paper count = 36 clusters.

Four Experimental Conditions: Every question family is evaluated across four parallel conditions (4 contexts per family):
1. supported_candidate: Contains the complete designated gold evidence passage(s) + corpus distractors (exactly 4 passages per context).
2. relevant_replacement: Essential evidence is replaced by a topically relevant passage from the same paper or topic that lacks the critical answer component (exactly 4 passages per context).
3. duplicate_replacement: Essential evidence is replaced by an exact duplicate copy of an existing distractor passage (exactly 4 passages per context).
4. natural_retrieval: Unconstrained end-to-end hybrid retrieval (BM25 + dense ChromaDB) reranked by BAAI/bge-reranker-base within the split corpus (allows up to 6 passages within a 2,000-token context budget).

Semantic Review Protocol: To ensure rigorous semantic ground truth, test annotations were adjudicated through a 12-checkpoint protocol where an assistant rater displayed and read complete, untruncated numbered passages for all 144 contexts. Every positive annotation was mechanically validated to contain exact verbatim quotation substrings from displayed passages (196 verified quotes, 0 failures). Negative annotations recorded specific missing information components. Label distribution across the 144 test contexts is: 69 sufficient (47.9%) and 75 insufficient (52.1%). In the controlled partition (108 contexts), insufficient prevalence is 66.7% (36 supported vs. 72 replacements). In natural retrieval (36 contexts), insufficient prevalence is 8.3% (33 sufficient vs. 3 insufficient).

5. Empirical Results & Findings

5.1 Primary Controlled Evaluation (N = 108 Contexts)

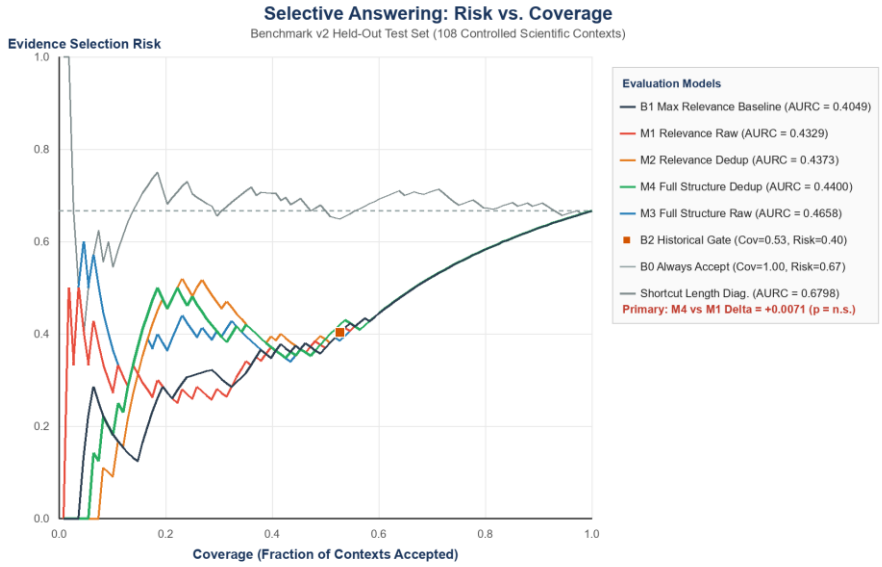


Figure 1: Risk vs. Coverage trade-off curves evaluated on the 108 controlled test contexts under uniform random tie ordering. M4 Full Structure Dedup (green line, discrete AURC = 0.4400) and M1 Relevance Raw (red line, discrete AURC = 0.4329) perform comparably (Delta = +0.0071, 95% bootstrap interval [-0.0503, +0.0679]). Baseline B1 Max Relevance (dark slate line) achieves the lowest discrete AURC (0.4049). B2 Historical Gate is plotted at its single discrete operating point (Coverage = 0.528, Risk = 0.404). Baseline B0 corresponds to the expected all-tied ranking AURC equal to the insufficient-label prevalence (Risk = 0.667), not a performance lower bound.

Table 1 reports the primary test results on the 108 controlled contexts (prevalence of insufficient contexts = 66.7%). Operating points reflect thresholds frozen on the validation set targeting coverages 0.50, 0.75, and 0.90. The length diagnostic achieved discrete AURC = 0.6798; this establishes only that this specific fitted length-based logistic regression diagnostic performed poorly, not that the benchmark is free of all possible shortcuts.

Model / Baseline	Discrete AURC	Cov@0.50 (Risk)	Cov@0.75 (Risk)	Cov@0.90 (Risk)
B0 (Always Accept)	0.6667	1.000 (0.667)	1.000 (0.667)	1.000 (0.667)
B1 (Max Relevance)	**0.4049**	0.315 (0.294)	0.472 (0.372)	0.852 (0.609)
B2 (Historical Gate)	N/A (Discrete)	— (Cov=0.528)	— (Risk=0.404)	T1=0.319, T2=0.056
M1 (Relevance Raw)	0.4329	0.250 (0.259)	0.491 (0.377)	0.824 (0.596)
M2 (Relevance Dedup)	0.4373	0.232 (0.520)	0.481 (0.385)	0.824 (0.596)
M3 (Full Structure Raw)	0.4658	0.352 (0.395)	0.491 (0.377)	0.907 (0.633)
M4 (Full Structure Dedup)	0.4400	0.222 (0.500)	0.546 (0.424)	0.722 (0.538)
Shortcut (Length Diag.)	0.6798	0.056 (0.500)	0.296 (0.656)	0.759 (0.683)

5.2 Paired Cluster Bootstrap Analysis (2,000 Resamples)

Table 2 presents the paired cluster-bootstrap uncertainty analysis on controlled contexts (36 paper clusters, random seed 20261104). Delta = Method A - Method B; negative values favor Method A.

Comparison Pair	Obs Delta	Boot Mean	Boot SE	95% Percentile CI	Verdict
M4 − M1 (Primary)	+0.0071	+0.0092	0.0305	[-0.0503, +0.0679]	**Not Supported**
M2 − M1 (Ablation 1)	+0.0044	+0.0052	0.0283	[-0.0520, +0.0603]	Contains 0
M3 − M1 (Ablation 2)	+0.0329	+0.0352	0.0228	[-0.0058, +0.0846]	Contains 0
M4 − M2 (Ablation 3)	+0.0027	+0.0041	0.0117	[-0.0199, +0.0270]	Contains 0
M4 − M3 (Ablation 4)	-0.0258	-0.0260	0.0334	[-0.0939, +0.0363]	Contains 0
B1 − M1 (Baseline)	-0.0281	-0.0253	0.0243	[-0.0668, +0.0267]	Contains 0
M4 − B1 (Method vs B1)	+0.0352	+0.0345	0.0318	[-0.0316, +0.0949]	Contains 0
Length − M4 (Shortcut)	+0.2398	+0.2410	0.0350	[+0.1722, +0.3081]	**Favors M4**

[PRIMARY HYPOTHESIS CONCLUSION] Primary Hypothesis Verdict: M4 − M1 discrete AURC difference is approximately +0.0071, with a 95% bootstrap percentile interval of [-0.0503, +0.0679]. The hypothesized improvement is NOT supported by the data. The interval includes differences favoring either method; this study does not establish equivalence. Baseline B1 (Max Relevance) achieves the lowest observed controlled-context AURC (0.4049).

5.3 Practical Generalization Check: Natural Retrieval Contexts (N = 36)

The natural retrieval partition evaluates performance in an unmanipulated end-to-end pipeline (BM25 + ChromaDB + bge-reranker-base). Only 3 of 36 contexts (8.3%) are semantically insufficient (ctx_te_0064, ctx_te_0112, ctx_te_0124).

Model / Baseline	Discrete AURC	Cov@0.50 (Risk)	Cov@0.75 (Risk)	Cov@0.90 (Risk)
B0 (Always Accept)	0.0833	1.000 (0.083)	1.000 (0.083)	1.000 (0.083)
B1 (Max Relevance)	0.0707	0.750 (0.074)	0.944 (0.059)	1.000 (0.083)
M1 (Relevance Raw)	0.0861	0.889 (0.062)	0.889 (0.062)	1.000 (0.083)
M2 (Relevance Dedup)	0.1037	0.972 (0.057)	0.972 (0.057)	1.000 (0.083)
M3 (Full Structure Raw)	0.0962	0.889 (0.062)	0.917 (0.061)	1.000 (0.083)
M4 (Full Structure Dedup)	**0.0501**	0.972 (0.057)	1.000 (0.083)	1.000 (0.083)
Shortcut (Length Diag.)	0.0644	1.000 (0.083)	1.000 (0.083)	1.000 (0.083)

Small-Sample Limitation: Caution Regarding Natural Retrieval Findings: Although M4 achieves the lowest observed AURC (0.0501) on the natural retrieval split, the partition contains only three insufficient contexts out of 36. Under paired cluster bootstrapping, the difference M4 − M1 has a 95% confidence interval of [-0.1176, +0.0160], which spans zero. Due to the very small negative sample count, this result cannot support claims of general superiority in practical retrieval settings.

5.4 Descriptive Subgroup Analysis: Question Complexity

Table 4 examines discrete AURC across question complexity typologies. Single-passage questions require extracting evidence from one passage; multi-passage questions require synthesizing evidence across distinct sections. Note that multi-passage questions comprise only 11 families (33 controlled contexts), making this an exploratory descriptive finding.

Subgroup	N	Insuff %	B1 MaxRel	M1 Raw	M2 Dedup	M4 Dedup
Controlled Single-Passage	75	66.7%	0.3166	0.3743	0.3472	0.3388
Controlled Multi-Passage	33	66.7%	0.5316	0.6894	0.3508	0.3803
All Single-Passage	100	50.0%	0.1617	0.1739	0.1662	0.1671
All Multi-Passage	44	56.8%	0.4214	0.4559	0.3343	0.2988

Methodological Nuance: Descriptive Multi-Passage Observation: On the 11 multi-passage families (33 controlled contexts), deduplicated models exhibit lower absolute discrete AURC values (M2 = 0.3508, M4 = 0.3803) compared to raw models (M1 = 0.6894, M3 = 0.6900). However, this is an 11-family descriptive observation and represents a difference in discrete AURC, NOT a '44.8% relative risk reduction' at a specified operational coverage. It does not establish causal claims about multi-passage reasoning or synthesis capabilities.

5.5 Duplicate-Invariance Property Test

Table 5 reports the observed absolute score perturbation |Score(C_dup) - Score(C)| when an exact duplicate of the highest-scoring passage is injected into all 36 supported_candidate test contexts post-selection.

Model / Baseline	Strictly Invariant?	Mean Abs Shift	Max Abs Shift	Std Shift
B1 (Max Relevance)	YES (Identical 0.0)	0.000000	0.000000	0.000000
M1 (Relevance Raw)	NO (Vulnerable)	0.318703	0.382319	0.038141
M2 (Relevance Dedup)	YES (Identical 0.0)	0.000000	0.000000	0.000000
M3 (Full Structure Raw)	NO (Vulnerable)	0.312393	0.387396	0.037412
M4 (Full Structure Dedup)	YES (Identical 0.0)	0.000000	0.000000	0.000000
Shortcut (Length Diag.)	NO (Vulnerable)	0.473078	0.478142	0.001642

Scope of Invariance: Duplicate Property Bounds: M2 and M4 exhibit exact zero score change specifically for the tested post-selection duplication operation on fixed contexts. This does NOT establish robustness to paraphrases, different metadata tags, re-retrieval pipeline changes, or evidence displaced by a strict token budget.

5.6 Operating Thresholds: Achieved vs. Descriptive Matched Coverage

Threshold Distinction: Deployable validation-frozen operating points must be strictly distinguished from descriptive matched-coverage points:

1. Deployable Frozen Operating Points: Operating thresholds T = {0.5518, 0.4923, 0.0414} for M1 and T = {0.5381, 0.2657, 0.0621} for M4 were frozen on the validation set targeting coverages 0.50, 0.75, and 0.90. On the held-out controlled test partition (N=108), M1 realized coverages of 0.250 (risk 0.259), 0.491 (risk 0.377), and 0.824 (risk 0.596). M4 realized coverages of 0.222 (risk 0.500), 0.546 (risk 0.424), and 0.722 (risk 0.538). The substantial deviations from target coverage reflect cross-split score distribution shifts.

2. Descriptive Matched-Coverage Points: When evaluating curves at exact integer ranks k = round(target * N) under uniform tie-breaking, M1 and M4 exhibit near-identical expected risks: at matched cov 0.50 (k=54), M1 risk=0.3889 vs M4 risk=0.3951; at matched cov 0.75 (k=81), both exhibit 0.5556 risk.

6. Threats to Validity & Methodological Disclosures

1. Same-AI Semantic Annotations: Test labels were generated through a 12-checkpoint untruncated AI assistant reading protocol with verbatim quotation verification. While every quotation was asserted against context text, labels are AI semantic annotations with prior domain exposure, not independent human judgments.
2. Model Freezing & Historical Exposure: All model parameters, scalers, and validation thresholds were completely frozen before the new v2 test contexts, questions, and annotations were constructed; however, we fully disclose historical test-paper exposure during an earlier invalid Stage 4 run prior to the v2 benchmark protocol.
3. Constructed Benchmark Nature: The controlled benchmark uses single-passage replacements (relevant and duplicate distractors), yielding a 66.7% synthetic insufficient prevalence, compared to 8.3% in natural retrieval.
4. Small Natural Negative Sample: The natural retrieval test split contains only 3 insufficient contexts (ctx_te_0064, ctx_te_0112, ctx_te_0124), yielding wide confidence intervals that span zero.
5. Shared Distractor Dependence: Shared distractor passages across question families can create residual inter-cluster dependence during bootstrap resampling.
6. Provenance Gap Documentation: A documented file re-copy occurred during checkpoint consolidation; the authoritative SHA-256 hash (fcd815fc...) was independently re-verified against raw passages.
7. Shortcut Bounds: The shortcut diagnostic achieves a discrete AURC of 0.6798; this poor performance establishes only that this specific fitted length-based diagnostic performed poorly, and does not prove the absence of all possible shortcuts or spurious correlations in the benchmark.
8. Zero Commercial API Spend: All experiments were executed locally on CUDA hardware using open-source models; commercial API spend is strictly $0.00.

7. Conclusion & Future Work

This project presents an empirical investigation of evidence-sufficiency gating in scientific question answering. Our primary hypothesis—that full deduplicated structural gating (M4) achieves lower discrete AURC than raw relevance gating (M1) on controlled scientific contexts—is NOT supported by the data (observed delta +0.0071, 95% bootstrap interval [-0.0503, +0.0679]). The interval includes differences favoring either method; this study does not establish equivalence. Simple maximum relevance (B1) achieves the lowest observed controlled-context AURC (0.4049) on the controlled benchmark.

Nonetheless, the empirical study yields two valuable architectural findings: (1) exact deduplication confers bounded duplicate invariance under specified operating assumptions (observed 0.000000 score change under post-selection duplication on fixed contexts), preventing the severe score perturbations observed in raw relevance models (>0.31 shift); and (2) on multi-passage questions, deduplication avoids the severe degradation observed in raw models (0.3803 vs 0.6894 discrete AURC). Future work should expand multi-passage family sizes, incorporate semantic paraphrase deduplication, and evaluate selective abstention directly within downstream generative synthesis.

8. References

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